Fluorescence Lifetime Analysis

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***Abstract*—This paper presents the design and implementation of a model for fluorescence emission. The fluorescence process is modeled using an exponential decay impulse response function (IRF), combined with a detector IRF that accounts for the detectors broadening effects. The system response is evaluated using pulsed excitation, and the resulting signal is sampled to emulate analog-to-digital conversion. Additionally, a heterodyne detection scheme is implemented to overcome sampling limitations by frequency shifting the fluorescence signal into a detectable band. Finally, fluorescence lifetime estimation is performed using both phase-based recovery and phasor analysis. The results show accurate recovery of the fluorescence lifetime under ideal conditions and highlight the impact of detector resolution and sampling constraints on signal reconstruction.**

***Index Terms*—fluorescence, fluorescence lifetime, sampling, phasor analysis**

I. Introduction

FLUORESCENCE is a process in which a fluorophore (a fluorescent chemical compound) emits light upon excitation. The main determining trait of the process is the fluorescence lifetime. This project simulates the fluorescence process with two systems of response with an excitation as input, and samples the output.

According to the background in the case study document, finding the fluorescence lifetime of a fluorophore is important to determine its chemical properties and can also be used to determine other properties like a material's changes in temperature, pH, ion concentration, protein interactions, and more.

We assume the measurement process for measuring emission collects many photons such that we are able to normalize the probability density of the fluorescence lifetime.

II. Background

We model our system using two impulse-response functions, which we convolve with our input $x(t)$. The first is the fluorescence emission IRF h_f in Equation 1.

$$h_f(t) = e^{-t/\tau}u(t), \tag{1}$$

where u is the unit step function and τ is the fluorescence lifetime of our function.

We call our second IRF the delay IRF, which models the probability density function (PDF) of average delay t_0 from when the fluorophore is first excited to when it emits light. We write the impulse of the PDF as in Equation 2.

$$h_d(t) = e^{-(t-t_0)^2/\sigma_t^2}u(t), \tag{2}$$

where σ_t is the variance around t_0.

With the IRFs, we multiply the signal by a modulation function $g(t)$ and then discretize according to some sampling rate. A visualization of our system is displayed in Fig. 1.

Taking the Fourier Transform $\mathcal{F}$ of the fluorescence IRF, we get the frequency response shown in Equations 3 and 4.

$$\mathcal{F}\{h_f(t)\} = \mathcal{F}\{e^{-t/\tau}u(t)\}, \tag{3}$$

$$H_f(j\omega) = \frac{1}{j\omega + \frac{1}{\tau}}. \tag{4}$$

Taking its magnitude, we should get a similar looking frequency response as in our model. This is noted for later comparison.

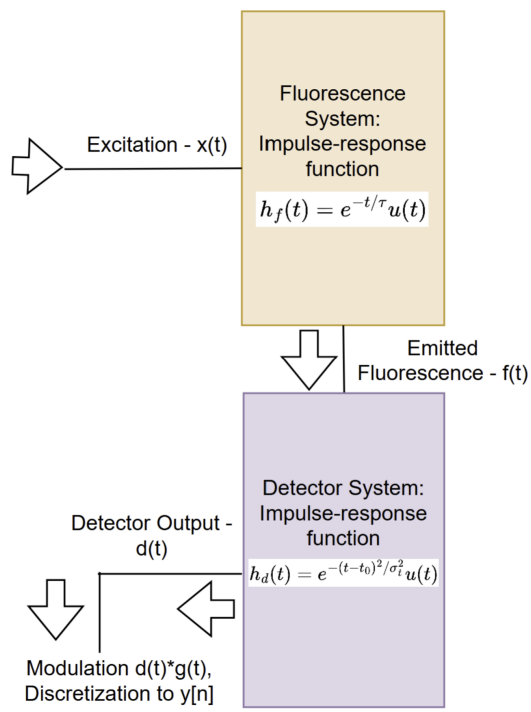

Fig. 1. Diagram of our system model. We put our excitation input through two IRFs, multiply by a modulation function (optional in some cases) and discretize to some sampling rate.

III. Methods

A. Fast time-domain detection

For fast time-domain detection, the excitation source was modeled as a period impulse train, representing a pulsed laser with a repetition period of $T_L = 12.5$ns. The continuous-time excitation signal $x(t)$ was defined as in Equation 5.

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_L). \tag{5}$$

This input was then convolved with the fluorescence emission IRF as defined in Equation 1 using a lifetime of $\tau = 1$ns. This output was then convolved with the detector IRF as defined in Equation 2 to model the PDF of the output to our system given this excitation source. This then gave us the signal $y(t)$ which was sampled, modeling the use of an analog-to-digital converter (ADC) with a sampling period $T_S = 0.2$ns. This is done to observe and model how different detector IRFs had an impact on the now digitized data $y[n]$ which we hope to closely resemble the emitted fluorescence $f(t)$. To do this, the variables in Equation 2, t_0 and σ_t, were varied and observed to find optimal values.

B. Heterodyne detection

In order to assess the importance of a proper sampling frequency for useful recovery of fluorescence lifetime information, assume that we are limited to a $f_s = 100$kHz sampling frequency. As noted in the background section, the fluorescence frequency response is given by Equation 4. Therefore, its magnitude and phase is given by Equations 6 and 7, respectively:

$$|H_f(j\omega)| = \frac{\tau}{\sqrt{1+(\omega\tau)^2}}, \tag{6}$$

$$\angle H_f(j\omega) = -\arctan(\omega\tau). \tag{7}$$

Now, assuming an input excitation x(t) as given by Equation 8 from the case study document:

$$x(t) = 1 + \cos(\omega_L t). \tag{8}$$

It is easy to note that the DC component will undergo a gain of τ and the cosine gets attenuated and shifted in phase from the magnitude and phase of $H_f(j\omega)$. Therefore, $f(t)$ has the form shown in Equation 9.

$$f(t) = \tau + \frac{\tau}{\sqrt{1+(\omega_L\tau)^2}}\cos(\omega_L t - \arctan(\omega_L\tau)). \tag{9}$$

Now, let the input $x(t)$'s cosine have a frequency of $f = 100$MHz as described in the case study assignment. Then, $\omega_L = 2\pi f$ rad/s.

The sampled signal is of the form $y[n] = f(nT_s)$, where T_s is the sampling period given by $T_s = \frac{1}{f_s} = 10\mu s$. Therefore, $y[n]$ is given by Equation 10. (Note that $f \times T_s = 1000$).

$$y[n] = \tau + \frac{\tau}{\sqrt{1+(\omega_L\tau)^2}}\cos(2\pi 1000n - \arctan(\omega_L\tau)). \tag{10}$$

Now, since n is an integer and cosine is an even function, $\cos(2\pi 1000n - \arctan(\omega_L\tau)) = \cos(\arctan(\omega_L\tau))$. Therefore, the expression for $y[n]$ does not depend on n and is a flat horizontal line for the given sampling and input frequency. Therefore, with a sampling frequency $f_s = 100$kHz in this example, it is not possible to recover useful fluorescence lifetime information.

However, if we are able to multiply $f(t)$ by some $g(t)$ such that we create a new, lower frequency, signal "copy" of $f(t)$, we should be able to properly sample this new signal with our relatively small sampling frequency of $f_s = 100$kHz. All that $g(t)$ has to do is shift the information to a detectable frequency, i.e. below the Nyquist frequency, which is half of our sampling frequency, yielding 50kHz. The simplest form of a $g(t)$ that let's us do that is shown in Equation 11, where $\omega_g = 2\pi f_g$.

$$g(t) = \cos(\omega_g t). \tag{11}$$

What $g(t)$ does in the frequency domain is the spectrum of $f(t)$ gets shifted by both f_g and $-f_g$, with the negative shift being what we are interested. This is because, if we choose f_g such that $f - f_g$ is less than the Nyquist frequency, we will be able to sample it. Therefore, $f - f_g \leq 50$kHz, yields $f_g \geq 99.95$MHz. To be safe, we chose $f_g = 99.99$MHz, which makes our final signal have a frequency component that is equal to 10kHz, a frequency we are able to sample properly.

To estimate the fluorescence lifetime from this shifted signal, we also need to copy the excitation signal. This should give us a signal to compare against the detected fluorescence signal. Let this signal be $c(t) = x(t)g(t)$. Then, its frequency component at 10kHz has no phase shift relative to the fluorescence. The post-multiplication by $g(t)$ low-frequency component of $y(t)$ has the form:

$$y(t) = \frac{M}{2}\cos((\omega_L - \omega_g)t - \phi), \tag{12}$$

where

$$M = \frac{1}{\sqrt{1+(\omega_L\tau)^2}}, \tag{13}$$

$$\phi = \arctan(\omega_L\tau). \tag{14}$$

Similarly, for $c(t)$:

$$c(t) = \frac{1}{2}\cos((\omega_L - \omega_g)t). \tag{15}$$

Therefore, if we compare the magnitude and phase of the 10kHz component of $y(t)$ and $c(t)$, we can recover the magnitude decrease and phase shift caused by fluorescence.

The modulation is computed as:

$$\Delta M = \frac{|Y(j\omega_L - \omega_g)|}{|C(j\omega_L - \omega_g)|}, \tag{16}$$

The phase shift is computed as:

$$\Delta\phi = \angle Y(j\omega_L - \omega_g) - \angle C(j\omega_L - \omega_g). \tag{17}$$

But, $\angle C(j\omega_L - \omega_g) = 0$ and $\angle Y(j\omega_L - \omega_g) = -\phi$. Finally, the fluorescence lifetime can be estimated from the phase from $\phi = \arctan(\omega_L\tau)$, which gives:

$$\tau = \frac{\tan(\phi)}{\omega_L}. \tag{18}$$

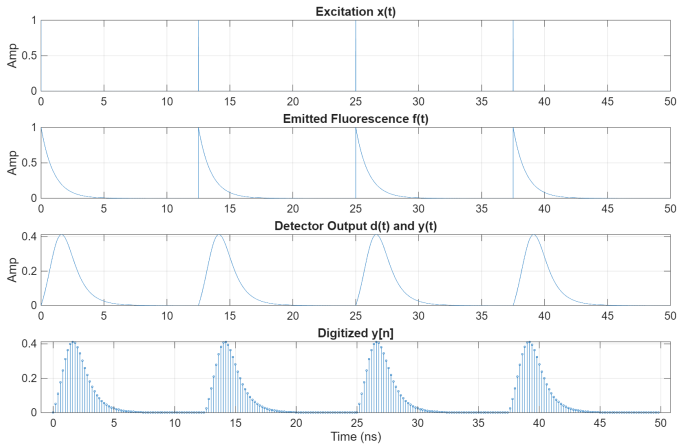

Fig. 2. Plot of $x(t), f(t), d(t), y(t), y[n]$ with $t_0 = 1$ and $\sigma_t = 1$.

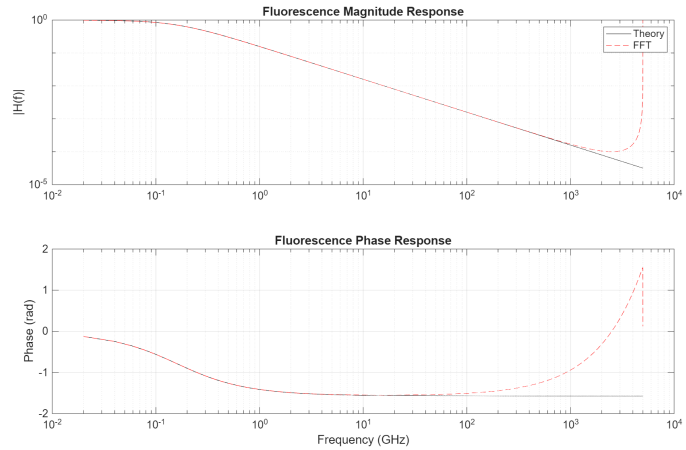

Fig. 3. Theoretical and Numerical plots of the frequency response of fluorescence.

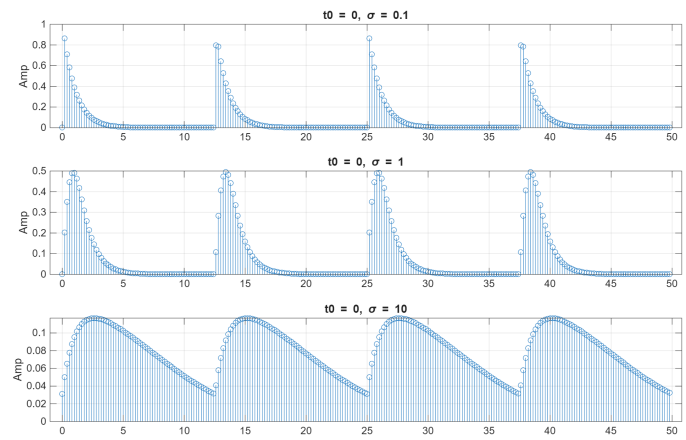

Fig. 4. Comparison of different detector IRF models.

C. Lifetime Phasor Analysis

In order to estimate the fluorescence lifetime and visualize it in 2D, we perform what is called phasor analysis. For any pixel p, we have the imaging decay function I_p. We get the components to plot by the quantities g and s, which we defined as in Equations 19 and 20:

$$g = \frac{\int_{<T_L>} I_p(t)\cos(\omega_L t)dt}{\int_{<T_L>} h_f(t)dt}, \tag{19}$$

$$s = \frac{\int_{<T_L>} I_p(t)\sin(\omega_L t)dt}{\int_{<T_L>} h_f(t)dt}. \tag{20}$$

After reshaping our .mat file into an array that can be indexed to a 1D decay function for each pixel, we find g and s for each decay function (I_p) and plot on the graph. We then estimate τ as follows:

$$\tau = \frac{s}{\omega_L g}. \tag{21}$$

The numerator integral, which we call a phasor transform, is closely analogous to the Fourier transform [1], and in a similar way has real/imaginary components. In the context of our system, we can see that the cosine term in our integral resembles a cosine series, which is the real part of the exponential transform of our function.

Therefore, g represents the real part of the phasor transform of our signal, while s represents the imaginary part of the phasor transform.

IV. Results

A. Fast time-domain detection

Fig. 2 shows the different time-domain signals defined within out system. As we would expect, the output every one of the signals is periodic with period $T = 12.5$ns since the input had this period.

Along with this, we can easily see how $d(t)$, the output from the detector, is like a spread out version of the emitted fluorescence $f(t)$. This is because our detector cannot measure the fluorescence instantaneously showing the physical realism of using a fast photomultiplier tube.

In this process the frequency response of our fluorescence IRF, $H_f(j\omega)$, found using MATLAB was also verified. This was done by calculating the theoretical frequency response, which was found earlier in Equation 4, and was then plotted in Fig. 3 against what was numerically found using MATLAB.

Both the theoretically and numerically found plots match extremely closely except for some behavior at much higher frequencies, likely due to numerical resolution limits and discretization effects in the MATLAB computation.

With this verified, different values of t_0 and σ_t were tested to see how it would impact the digitized data $y[n]$. This parameter sweep can be seen in Fig. 4 where different values of the time delay and variance were tested to determine how the detectors parameters affect the digitized data.

The different values of σ_t were chosen to represent a range of detector performances. As seen in Fig. 4, when $\sigma_t = 0.1$, the detector output closely matches the emitted fluorescence shown in Fig. 2, which would be the ideal output for the detector. This behavior is expected because the detector IRF is modeled as a Gaussian distribution with variance σ_t. As the variance approaches zero, the Gaussian increasingly resembles an impulse function. If the detector IRF were a perfect impulse, the detector output would exactly match the emitted fluorescence signal. While this is not physically achievable,

the results demonstrate that even a relatively small variance of $\sigma_t = 0.1$ produces a response that closely approximates the ideal case.

As σ_t increases, however, this impulse approximation becomes less accurate. When $\sigma_t = 1$, the detected signal still retains some resemblance to the emitted fluorescence, though noticeable broadening begins to appear. For $\sigma_t = 10$, the detector response becomes significantly broadened, and the measured signal is heavily smeared in time. This illustrates how poorer detector temporal resolution can distort the measured fluorescence signal, making it more difficult to accurately recover the original emission profile.

B. Heterodyne detection

Fig. 5 shows the original excitation input $x(t)$ and the fluorescence signal $f(t)$. As expected, the fluorescence signal has smaller amplitude and a phase delay due to its lifetime.

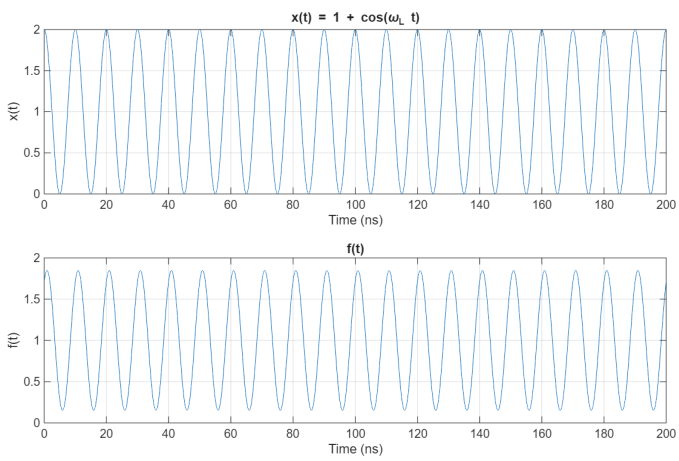

Fig. 5. Original $x(t)$ and fluorescence signal $f(t)$.

Fig. 6 shows the signal $g(t)$ described in the methods section and the product of it with the fluorescence signal.

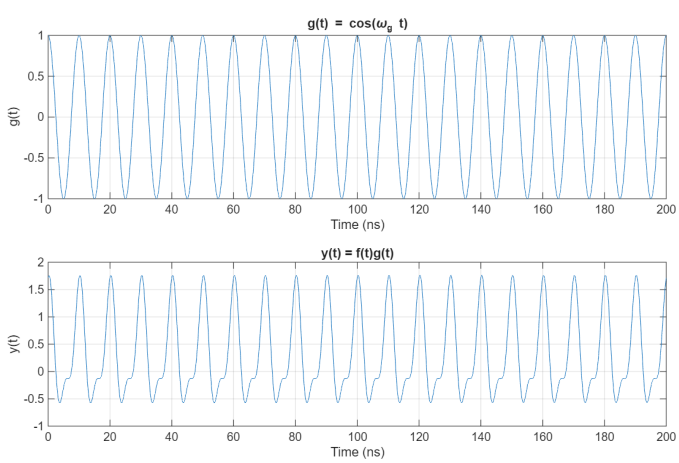

Fig. 6. Plot of $g(t)$ used for frequency shifting and $y(t)$ as the result of the multiplication of it by the fluorescence signal.

In order to see if what we showed in the methods section that choosing $f_g = 99.99$MHz would create the component at

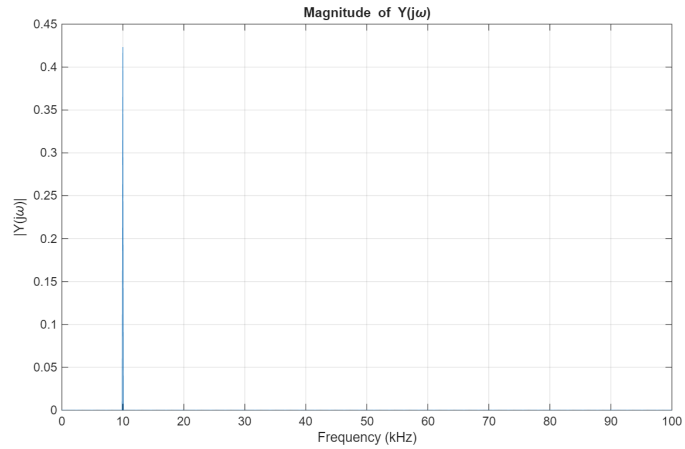

Fig. 7. Magnitude of the resulting $Y(j\omega)$ after proper frequency shifting for sampling.

10kHz we would be able to sample, Fig. 7 shows exactly this frequency component we were expecting.

To recover the fluorescence lifetime, we multiplied both the fluorescence signal and the copy of the excitation signal using the same $g(t)$.
As reported by MATLAB, the magnitude ratio between the fluorescence and copy signals yielded:

$$\Delta M = \frac{|Y(j\omega_L - \omega_g)|}{|C(j\omega_L - \omega_g)|} \approx 0.8467, \tag{22}$$

and the phase shift reported was:

$$\Delta\phi = \angle Y(j\omega_L - \omega_g) - \angle C(j\omega_L - \omega_g) \approx -0.561. \tag{23}$$

From that, as shown in the methods section, we can recover the fluorescence lifetime by the phase shift, which MATLAB reported as $\tau = 1.000$ns, which is exactly the value of τ we used in the simulation.

C. Phasor Plots

We plot a general fluorescence lifetime on Fig. 8. We can see that the lifetime, or when the amplitude drops to about $1/e$, is on the scale of about 1ns.

Fig. 9 gives us all phasor transform values for our data.

Our image pixel values are primarily in the Bound NADPH and Bound NADH regions according to the classifications in the case study document. We see that the values of τ are relatively high, indicating a longer emission decay.

Generally, we expect a lifetime of 1 ns, meaning that our values are slightly higher than we would expect, most of which are between 1 and 2. However, we are within reasonable magnitude of the true lifetime using this method.

V. Conclusion

In this case study, we developed and analyzed a simulation for fluorescence lifetime measurement using both time-domain and frequency-domain approaches. By modeling the fluorescence signal and detector response, we showed how detector resolution and sampling constraints can significantly affect the quality of the measured signal.

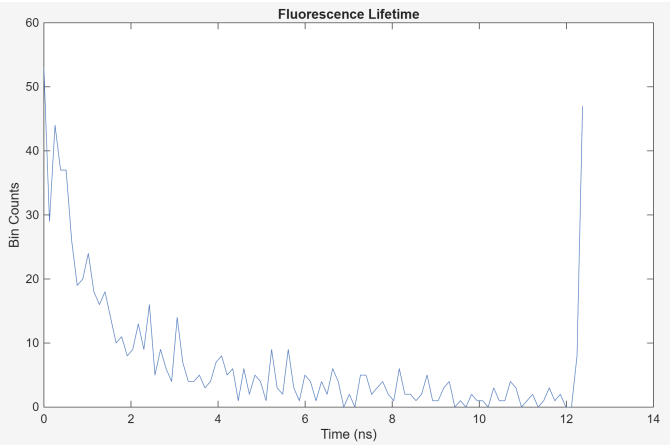

Fig. 8. Plot of the fluorescence lifetime by aggregating the counts every 100 pixels.

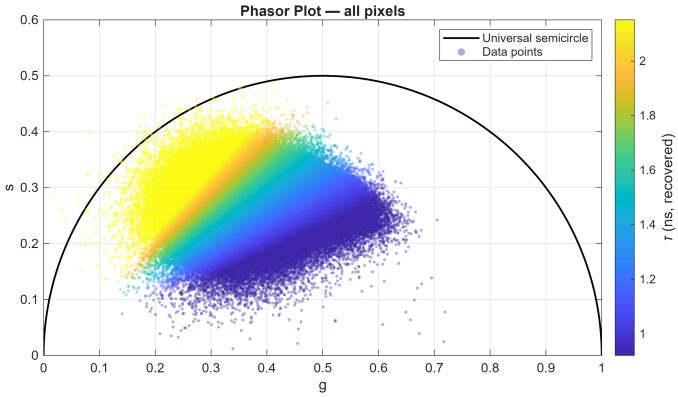

Fig. 9. Phasor plot for the fluorescence spectra of our data.

The time-domain results highlighted that better detector precision leads to a closer match between the measured and true fluorescence signal, while poorer resolution introduces noticeable distortion. To address sampling limitations, a heterodyne detection method was used to shift the signal into a lower frequency range, allowing accurate recovery of the fluorescence lifetime even with limited sampling rates.

Finally, phasor analysis provided a simple and intuitive way to visualize and estimate fluorescence lifetimes from the simulated data. Overall, the results demonstrate that fluorescence lifetime information can be reliably extracted using a combination of signal processing techniques, even under practical measurement constraints.

Acknowledgment

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References

[1] R. A. L. Schoop et al., “Phasor theory of fluorescence lifetime imaging utilized on a maximum range of frequencies for prostate tissue analysis,” Biomed Opt Express, vol. 16, no. 11, pp. 4885–4901, Oct. 2025, doi: 10.1364/BOE.575841.